

packages/graph-view/src/utils/filters.ts

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Overview

`packages/graph-view/src/utils/filters.ts` introduces typed filtering utilities for graph nodes and edges. The module replaces ad-hoc logic with reusable functions that operate on pre-computed indices.

Key changes

- **Imports** (lines 1-3):

```
``ts import type { GraphNode, GraphEdge } from "@novadiff/graph-core/types"; import type { FilterState, NodeType, Complexity, EdgeCategory } from "../store"; import { EDGE_CATEGORY_MAP } from "../store";
```
- `filterNodes` (lines 19-52):
 - Filters by `nodeTypes` and `complexities`.
 - Uses `nodeIdToLayerIds: Map<string, Set<string>>` for O(1) layer membership checks, matching the "any-layer-wins" semantics described in the comment.
- `filterEdges` (lines 57-76):
 - Keeps only edges whose `source` and `target` are in `visibleNodeIds`.
 - Filters by edge category via `getEdgeCategory`.
- `getEdgeCategory` (lines 81-88):
 - Looks up an edge's category in `EDGE_CATEGORY_MAP`; returns `null` if the type is unknown.

Impact

- **Performance:** The comment in `filterNodes` notes that the new O(1) layer lookup removes the former $O(N \times L \times K)$ cost that dominated export time on large graphs (#102).
- **Correctness:** Explicit type checks prevent accidental inclusion of unsupported node types or complexities.
- **Maintainability:** Centralized filtering logic simplifies future extensions (e.g., new node/edge attributes).

Risks & follow-ups

- **Edge category coverage:** `getEdgeCategory` returns `null` for unknown types, causing `filterEdges` to drop those edges. Verify that `EDGE_CATEGORY_MAP` contains all used edge types.

- **Layer index consistency:** `nodeIdToLayerIds` must be correctly populated in the store; stale indices could misfilter nodes.
- **Compatibility:** Confirm that existing consumers still receive the same node/edge sets after the refactor (unknown from the diff).
- **Testing:** Add unit tests for `filterNodes` and `filterEdges` covering cases such as no layers selected, multi-layer nodes, and unknown categories.